Saumya B

Bengaluru, India · Email · LinkedIn · Website · Github

Education

Georgia Institute of Technology — Atlanta, USA
M.S. in Electrical and Computer Engineering *(Deferred)* — Fall 2027

M S Ramaiah Institute of Technology — Bangalore, India
B.E in Electronics and Communication Engineering *GPA: 8.72* — 2023

MES BRS PU College
12th grade, PCMB | *Score: 95.86%* | **College topper** — 2019

PoornaPrajna High School
10th grade | *Score: 96.33%* | **State 10th rank** — 2017

Patents & Publications

[A1] An Empirical Study on MC Dropout–Based Uncertainty–Error Correlation in 2D Brain Tumor Segmentation [arXiv]

[A2] Reproducible Evaluation of Data Augmentation and Loss Functions for Brain Tumor Segmentation [arXiv]

[P1] System and method for controlling laser scanning subsystems in a multi-laser control system
Indian Patent Office, Application No. 202641114431, Filed Sept 2026

[C1] Smart shoes for Parkinson's Patients
2023 IEEE World Conference on Communication & Computing (WCONF) **[DOI]**

Work Experience

EmSys Lab, Indian Institute of Science (IISc) — Bangalore, India
Project Associate — June 2024 – Present

- **Feature Selection Library: FeS-Lib** - Currently building an open-source feature selection library for anomaly detection in electric vehicles (sponsored by ***Schaeffler, Germany***)
- **Security Event Anomaly Detection Pipeline**
  - Built a real-time ETL pipeline in Python that captures live Windows kernel events and streams them into a dockerized PostgreSQL graph-based schema (subject, file, netflow tables with an event edge table), sustaining 3000 events/min with deduplication logic using hash-based entity lookups
  - Designed data preprocessing and feature-engineering workflows to transform heterogeneous security logs into structured tabular data for anomaly detection
  - Built a separate ingestion pipeline to parse static Sysmon log datasets (35,000+ events) into the same graph-based schema, and generated ground-truth labels to evaluate anomaly detection accuracy
  - Implemented a graph-based anomaly detection model to score live security data for intrusion detection
- **Control software and dashboard development for 3D printer** — Developed a real-time 3D printer control and monitoring application in Python, integrating a graphical dashboard, asynchronous processing, and hardware/laser control; reduced inter-layer delays by 78% (sponsored by ***Ministry of Heavy Industries, Government of India***)

ML Lab, Indian Institute of Science (IISc) — Bangalore
Research Intern — Jul 2023 – Oct 2023

- Implemented classical computer vision techniques (Canny edge detection, Hough transform) for feature extraction and lane detection, and deployed the resulting path-detection and route planning modules on Raspberry-Pi for real-time inference.
- Managed data collection, logging, and preprocessing of real-world experimental data to support downstream analysis and model training.

Lenovo Private Limited — Bangalore
Graduate Engineer Trainee — Jan 2024 – May 2024

- Gained practical experience in Linux and Windows system administration, including user management, permissions, and system services.

Projects

Uncertainty-Error Correlation in Brain Tumor Segmentation [*arXiV*]
- Sole-authored arXiv preprint showing that MC Dropout variance is an unreliable error signal in U-Net brain tumor segmentation (global r ≈ 0.3, boundary $|r| < 0.05$), with the result consistent across four augmentation regimes and supported by paired significance testing. Implemented U-Net from scratch using first principles.

AI Goggles - Live Video Captioning for the Visually Impaired [*Github*]

◦ Engineered an edge-deployable assistive captioning system on a Raspberry Pi 4: replaced a non-exportable BLIP model with a ViT+GPT-2 encoder-decoder chosen for modular ONNX export, patched GPT-2's KV-cache path to make the decoder traceable, and ran CPU inference with spoken output.

Pneumonia Detection from Chest X-Rays – Comparative Study [*Github*]

◦ Ran a systematic comparative analysis of 5+ CNN training strategies for pneumonia detection from chest X-rays, with Grad-CAM interpretability analysis

Assistive Device for Parkinson's Patients **Best Project Award**

◦ Developed a real-time biomechanical signal-processing pipeline for Freeze of Gait detection on embedded hardware using FSR sensors and a pressure-ratio threshold classifier

Skills

Languages	Python, C, C++, HTML5, CSS, JavaScript
ML & Frameworks	PyTorch, TensorFlow, Scikit-Learn, Keras, ONNX Runtime
Computer Vision	OpenCV, PIL, scikit-image
Data & Development	Numpy, Pandas, Git, FastAPI, PyQt5, Docker
Systems & Concurrency	Multi-threading, asynchronous processing, Qt/PyQt signal–slot architecture, sockets (TCP/UDP)